IMPLEMENTATION OF FULL ADDER USING 3:8DECODER

DESIGN

```
module fa(s ,c,a,b,cin);
output s,c;
input a,b,cin;
wire  y0,y1,y2,y3,y4,y5,y6,y7;
wire r1,r2,r3,w1,w2,w3,w4,w5,w6,w7,w8;
//gate level modeling
not t1(r1,a);
not t2(r2,b);
not t3(r3,cin);
and t4(y0,r1,r2,r3);
and t5(y1,r1,r2,cin);
and t6(y2,r1,b,r3);
and t7(y3,r1,b,cin);
and t8(y4,a,r2,r3);
and t9(y5,a,r2,cin);
and t10(y6,a,b,r3);
and t11(y7,a,b,cin);
xor t12(s,y1,y2,y4,y7);
xor t13(c,y3,y5,y6,y7);
endmodule
```

TESTBENCH

```
module fa_tb();
wire s,c;
reg a,b,cin;
fa uut(s,c,a,b,cin);
initial begin
```

```
a=0; b=0; cin=0;

#100; a=0; b=0; cin=1;

#100; a=0; b=1; cin=0;

#100; a=0; b=1; cin=1;

#100; a=1; b=0; cin=0;

#100; a=1; b=0; cin=1;

#100; a=1; b=1; cin=0;

#100; a=1; b=1; cin=1;

end


endmodule
```

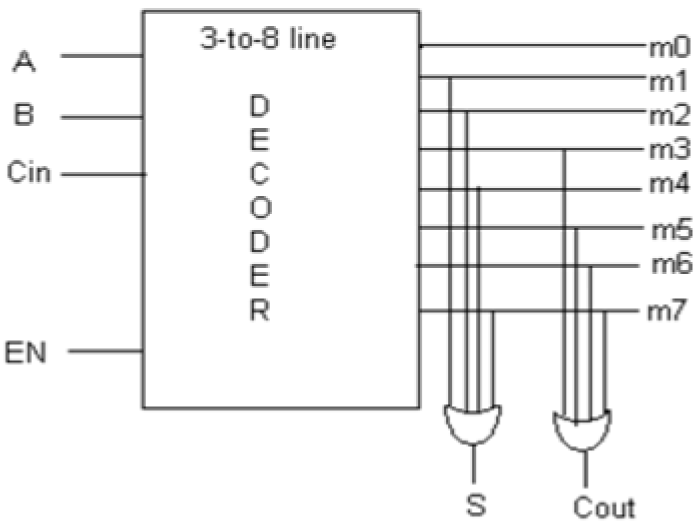
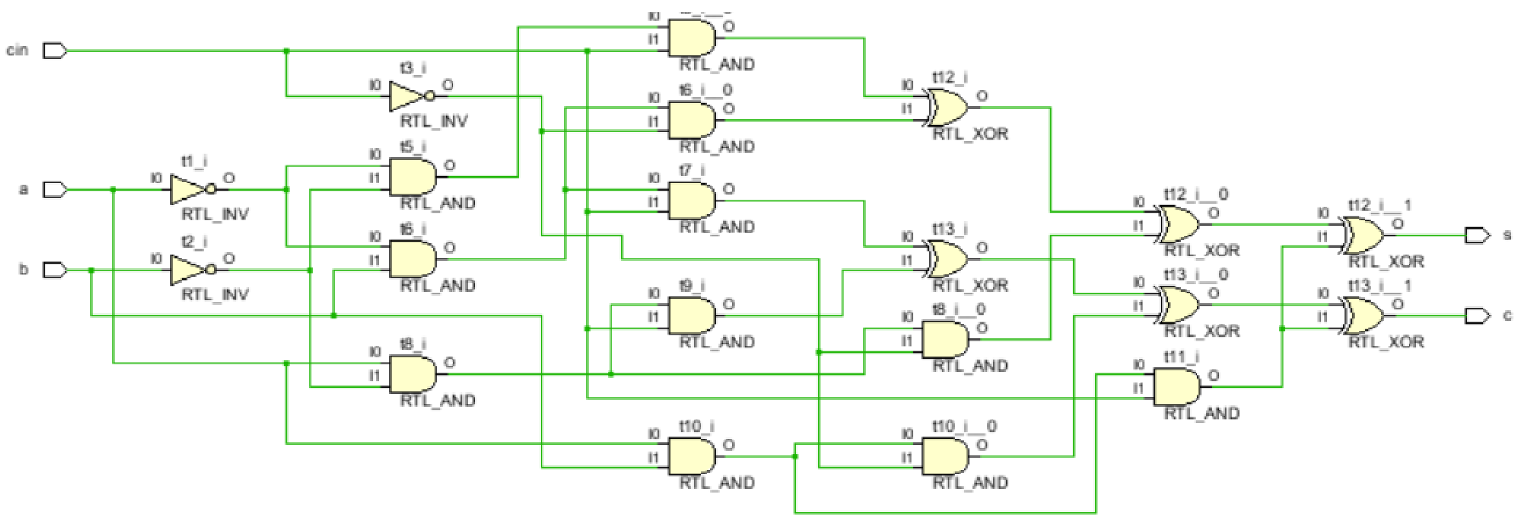
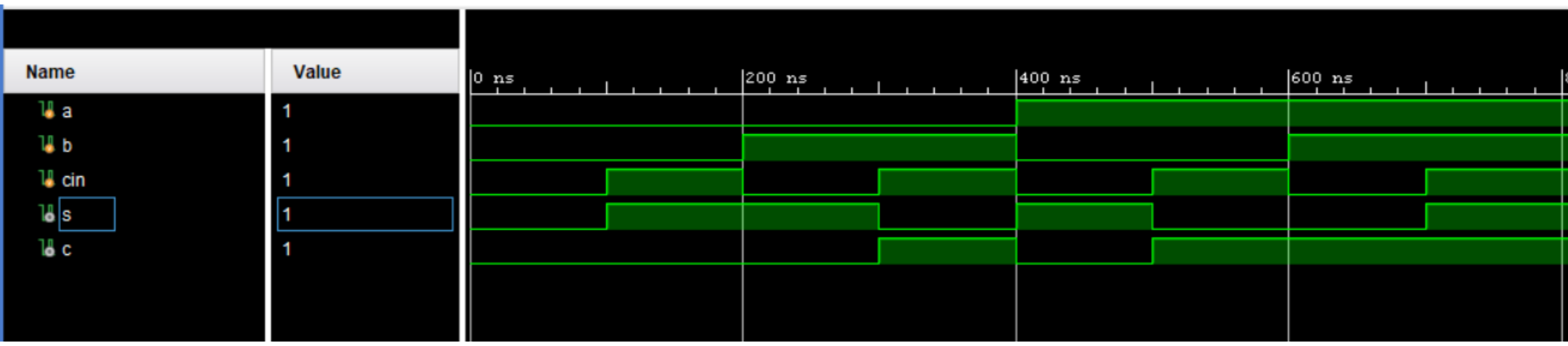


Fig1: Full Adder Implementation using 3:8 decoder





TRUTH TABLE

A	B	Cin	Sum (S)	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1